Persistent Industrial Thermal Source Detection

Model summary, work completed, results, limitations, and next steps

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1. Executive summary

The project detects persistent industrial thermal sources from NASA FIRMS active-fire observations and satellite context. The final selected ranking model combines a compact LightGBM ensemble with a guarded computer-vision residual based on Sentinel-2 imagery. The selected branch is called **guarded_cv_tabular**.

On the confirmatory six-country panel, the selected model achieved macro country PR-AUC 0.929336, macro country F1 0.816620, and macro country ROC-AUC 0.948094. The protected compact baseline achieved 0.905404 PR-AUC, 0.766876 F1, and 0.919853 ROC-AUC. The guarded model therefore improved PR-AUC by 0.023932 and F1 by 0.049744.

The ranking model is frozen. The deployment threshold is not frozen because the reported F1 uses held-country validation thresholds, not one threshold calibrated for India. India was not loaded during model selection and remains the final holdout.

Main result

Model	Macro F1	Macro PR-AUC	Macro ROC-AUC	Worst-country PR-AUC
Guarded CV + tabular	0.816620	0.929336	0.948094	0.840037
Compact baseline	0.766876	0.905404	0.919853	0.825015
All77 diagnostic	0.734732	0.893485	0.922493	0.848884

F1 values use country-specific median fold-local leave-one-country-out thresholds. They are not results from one deployment threshold.

2. Data and preprocessing

The raw dataset contains 78 FIRMS CSV files covering 19,342,072 detections across seven countries from 2019 to 2024. The audit found no null rows, malformed dates, or exact duplicate rows. The countries are Algeria, Angola, India, Indonesia, Iraq, Libya, and Nigeria.

Source construction

The unit of analysis is a source rather than an individual detection. Detections were mapped to a 1 km metric grid. For example, Libya was reduced from 427,517 detections to 4,824 sources, approximately 89 detections per source. The 1 km setting retained about 88 to 92 percent EOG flare recall in the clustering study.

Validation groups use 10 km spatial blocks. This prevents fragments of the same physical flare from appearing on both sides of a train-validation split. A 2022 to 2024 common window was used for cross-country modelling so different year coverage could not act as a country identifier.

Supervision

EOG flare inventory matches provide known gas-flare positives. The four originally labelled training countries contained 1,342 active flare inventory sites. An unmatched FIRMS source is not necessarily negative because kilns, cement plants, steel facilities, and other industrial sources can be absent from the EOG gas-flare inventory. The learning problem is therefore treated as positive-unlabeled learning.

Confirmatory panel

Country	Sources	EOG positives
Algeria	48	15
Angola	50	6
Indonesia	48	17
Iraq	48	18
Libya	51	15
Nigeria	49	16
Total	294	87

This is an enriched evaluation panel. Its precision, F1, and positive rate are not estimates of deployment prevalence across all sources.

3. Features and model development

FIRMS features

The source-level FIRMS feature set captures persistence, active days and months, detection frequency, fire radiative power, brightness statistics, MIR minus LWIR temperature contrast, sensor saturation, seasonality, local solar time, spatial spread, and safe cross-instrument information. Latitude, longitude, country, NASA type, and EOG distance were excluded from learned features. Raw NOAA-20 availability was also controlled because NOAA-20 was missing for Angola and Indonesia and could otherwise become a country proxy.

Development sequence

- Early Libya and Algeria baselines: Isolation Forest F1 0.340, rule baseline F1 0.633, and LightGBM F1 0.691 with PR-AUC 0.740 and ROC-AUC 0.930.
- Common-window and leave-one-country-out work showed that regional background and country imbalance mattered more than adding a temporal neural network.
- A compact multimodal model added selected Sentinel-2 and WorldCover summaries to the FIRMS features.
- The guarded computer-vision branch added a bounded visual correction rather than replacing the stable compact score.

The early metrics and final metrics use different cohorts and validation protocols. They describe model development and should not be compared as one continuous leaderboard.

Final compact branch

The compact branch uses 69 selected FIRMS, Sentinel-2 summary, and WorldCover columns. The frozen package contains nine LightGBM models: three models for each confirmatory seed 272, 273, and 274.

Final visual branch

The visual branch uses a pinned Sentinel-2 RGB MoCo ResNet-18 encoder without fine-tuning. It processes a full 2 km view and a center 1 km view. Identity, horizontal flip, vertical flip, and 90 degree rotation predictions are averaged. The two views produce a combined 1,024-dimensional embedding. The model also uses 77 image and land-cover features and 41 morphology features.

Within each training fold, the visual representation is normalized, standardized, reduced to 12 principal components, and fitted with five-bag regularized logistic positive-unlabeled probes. There is one final visual model for each confirmatory seed.

4. Final fusion and validation design

Fusion

For each seed, the three compact model probabilities are averaged. Compact and visual scores are calibrated from foreign out-of-fold references. The guarded correction uses alpha 0.5 for every confirmatory seed. The exact rule is:

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fused = tabular + alpha * 4 * tabular * (1 - tabular) * (visual - tabular)
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The uncertainty term 4 * tabular * (1 - tabular) makes the visual correction strongest when the compact score is uncertain and weak when the compact score is already near 0 or 1. The final ranking score is the mean of the three seed-specific fused scores. If dense imagery is missing, the defined fallback is the mean of the three calibrated compact pipelines.

Validation controls

- Outer leave-one-country-out evaluation across Algeria, Angola, Indonesia, Iraq, Libya, and Nigeria.
- Every scaler, PCA transform, visual probe, LightGBM model, fusion alpha, and validation threshold is fitted without the held country.
- Discovery seed 271 is excluded from confirmation. Fresh seeds 272, 273, and 274 were fixed before the run.
- India was never loaded during selection or confirmation.
- The all77 and nested-selector branches were diagnostic only and could not change the promotion decision.
- The paired uncertainty analysis used 5,000 positive-stratified 10 km block bootstrap repeats with fixed seed 12012.

Acceptance result

All 12 preregistered acceptance conditions passed. These included minimum PR-AUC gain, minimum number of improved countries, limits on the worst country and Indonesia losses, F1 non-inferiority, a bootstrap lower bound, review-budget limits, positive gains for all fresh seeds, a median seed-gain requirement, and nonzero fusion weights for at least two seeds.

The paired bootstrap mean PR-AUC gain was +0.023764 with a 95 percent interval from -0.000632 to +0.051502. The interval crosses zero. The result passes the prespecified non-inferiority lower guard of -0.010, but it is not conventional positive statistical significance.

5. Detailed results

Country metrics

Country	Compact PR-AUC	Guarded PR-AUC	PR-AUC change	Compact F1	Guarded F1
Algeria	0.825015	0.840037	+0.015022	0.666667	0.714286
Angola	0.958333	1.000000	+0.041667	0.909091	0.923077
Indonesia	0.955917	0.950863	-0.005054	0.692308	0.857143
Iraq	0.877719	0.908707	+0.030988	0.848485	0.875000
Libya	0.909849	0.925075	+0.015226	0.764706	0.789474
Nigeria	0.905589	0.951333	+0.045744	0.720000	0.740741

Five of six countries improved in PR-AUC. Indonesia declined by 0.005054, which remained inside the fixed country-loss guard of 0.020. Angola PR-AUC 1.000 is based on only six known positives and should not be interpreted as population perfection. F1 again uses country-specific held-country thresholds.

Seed stability

Seed	Compact PR-AUC	Guarded PR-AUC	PR-AUC gain	F1 gain	Alpha
272	0.906547	0.929960	+0.023413	+0.051996	0.5
273	0.905569	0.928411	+0.022842	+0.047596	0.5
274	0.908584	0.930390	+0.021806	+0.053570	0.5

All three fresh seeds produced positive PR-AUC gains. This measures algorithmic seed stability on the same panel. It does not measure new-country or new-cohort variance.

Top 20 percent review budget

Country	Sources reviewed	Compact positives found	Guarded positives found	Change
Algeria	10	8	8	0
Angola	10	6	6	0
Indonesia	10	10	10	0
Iraq	10	10	10	0
Libya	11	11	10	-1
Nigeria	10	10	10	0
Total	61	55	54	-1

At this operating slice, guarded recovered 54 of 87 known positives versus 55 for compact. The only loss was one positive in Libya. The ranking metric improved, but this particular review budget did not improve.

6. Frozen model, limitations, and next work

Frozen items

- Selected branch: guarded_cv_tabular.
- Confirmatory protocol: 12b-guarded-confirmatory-ensemble-v1.
- Source model commit: 288de63.
- Seeds: 272, 273, and 274. Fusion alpha: 0.5 for every seed.
- Artifacts: nine compact LightGBM models, three visual PU models, calibration references, selected schema, verified embedding and morphology caches, and exact score construction.
- Missing dense-imagery fallback: mean of three seed-specific calibrated compact pipelines.

No additional tuning on the six-country foreign panel should be performed under this model version. Any material change should create a new protocol and model version.

Important limitations

- The 294-source panel is enriched and was reused for confirmation. It is not a population precision study.
- Three seeds test algorithmic variance only.
- EOG labels gas flares, not every industrial thermal source. Unmatched sources are not verified negatives.
- The bootstrap interval crosses zero.
- Only 294 sources entered the final dense-CV QA cohort, so visual coverage and selection risk remain.
- India imagery and labels were not loaded. No India score exists yet.
- The pilot threshold 0.616443 is not deployment calibrated and must not be used as a final cutoff.

Next controlled step

First, calibrate a fixed threshold or review budget from foreign out-of-fold predictions only. Second, obtain the matching India compact imagery summaries and dense Sentinel-2 chips. Third, lock code, hashes, operating policy, and output schema. Finally, run India once and report the result without model retuning. For cost control, use the compact model to prefilter India sources and apply the dense visual reranker only to the bounded candidate set.

Current status

Ranking model: frozen and selected.
Selected model: guarded_cv_tabular.
Deployment threshold: pending calibration.
Independent holdout: India remains sealed.